COURSE WORK 02- GROUP ASSIGNMENT SUBMISSION

PRINCIPLES OF DATA SCIENCE

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Coventry University- MSc Data Science

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Github Link: https://github.com/Group-Courseworks-MScDS/fashion_retail_demand_analysis.git

Contents

1. **Introduction** 1
2. **Methodology** 2
3. **Data Set and Data Collection** 2
4. **Data Cleaning and Preprocessing** 3
5. **Exploratory Data Analysis (EDA)** 4
6. **Clustering** 11
   - 6.1. **Preprocessing** 11
     - 6.1.1. **Numerical Feature Preprocessing** 11
     - 6.1.2. **Categorical Feature Preprocessing** 11
   - 6.2. **Optimal Number of Clusters (Elbow Method)** 12
   - 6.3. **Cluster Validation Using Silhouette Score** 12
   - 6.4. **Summarize** 13
   - 6.5. **Visualizing** 13
   - 6.6. **Business Insights from Clustering** 15
7. **Predictive Findings and Business Outcomes** 16
   - 7.1. **Significant Findings** 16
     - 7.1.1. **Model Performance and Reliability** 16
     - 7.1.2. **Impact of Promotion** 17
     - 7.1.3. **Seasonal Influence** 17
   - 7.2. **Business Outcomes and Recommendations** 18
   - 7.3. **Conclusion** 18

1. Introduction

This group project focuses on carrying out an in depth data analysis of a real-world fashion retail dataset with the intention of supporting better, data-driven decisions that improve both operational efficiency and customer satisfaction. At its core, the project focuses on a common business challenge of understanding how sales perform, how customer demand changes, and how different products behave so that retail operations and planning can be optimized more effectively.

By working with the dataset, the said analysis aims to uncover important trends, group products into meaningful categories, and build a predictive model that can estimate future sales under different business conditions. To achieve this, several techniques are used, including data cleaning, data exploratory analysis, clustering, and machine learning based prediction. The project also looks closely at factors such as pricing strategies, the effects of promotions, seasonal demand patterns, and overall product performance. In addition, forecasting methods are applied to predict future demand, helping businesses make smarter decisions around inventory management, marketing efforts, and resource allocation.

Overall, this group work shows how data driven approaches can significantly improve decision making in the retail industry. This report presents the dataset, methods, results, and key insights in a clear and organized way, while also discussing the challenges faced and possible directions for future improvement.

2. Methodology

The data set selection, exploratory data analysis steps, problem solving steps and model section will be discussed in detail under this section. During the discussion the factors considered for the decision making process will also be discussed.

3. Data Set and Data Collection

The dataset used in this project was downloaded from a publicly available fashion retail dataset which was provided in a structured CSV format. Since the data had already been collected from real world retail operations, there was no need for primary data collection methods such as surveys or experiments. Therefore, the main focus was on extracting the dataset, checking its structure, and preparing it for analysis using a full Extract, Transform, Load (ETL) process. This process involved dealing with missing values, fixing any inconsistencies, and transforming both categorical and numerical data into a usable format. In summary, these steps were conducted to ensure that the dataset was clean, reliable, and ready for further analysis, including tasks like clustering and predictive modeling.

The dataset was chosen for this project as it is well suited for the main objective of the analysis. Because this data set includes important retail related features such as price, promotional indicators, seasonal factors, product categories, materials, origin, and sales volume. These variables play a major role in shaping customer demand and overall business performance. Therefore, it made this dataset highly relevant for studying sales trends and customer behavior. In addition, the combination of both numerical and categorical data allowed for the use and experiment with a of a wide range of analytical methods. This includes techniques like clustering for grouping similar products and machine learning models for predicting future demand. In a nutshell, the dataset provides a solid foundation for generating meaningful insights and supporting effective, data driven decision making in a retail environment.

4. Data Cleaning and Preprocessing

In this project, several data preprocessing steps had to be taken to make sure the dataset was clean, consistent, and ready for analysis and modeling. As the first step, the process began with handling missing values. For numerical features like price, missing values were replaced using the median to minimize the effect of extreme values. For categorical features, the most common category was used. This approach helps keep the data reliable without introducing unnecessary bias.

The second step was to clean the data while checking for consistency. This included removing duplicate records, fixing data type issues, and standardizing categorical values such as season, section, and material. Performing this step, it was targeted to ensure that similar categories were grouped correctly rather than being treated as separate entries. Outliers were addressed, by giving special focus on the variables like price and sales volume, using methods such as the Interquartile Range (IQR). This was important to prevent extreme values from skewing the results of clustering and prediction models.

Furthermore, the Product Category, Brand, and Currency columns were removed because each of these fields contained the same values for every record in the dataset. Therefore, these columns does not provide meaningful information for analysis, clustering, and prediction. Performing this, it reduce the redundancy and makes dataset more efficient for analyzing, clustering, and forecasting.

In addition, feature transformation and encoding were applied to prepare the data for modeling. Numerical variables were standardized so that all features had a similar scale, which is particularly important for algorithms like K-Means clustering. Categorical variables were converted using one-hot encoding, making them usable in machine learning models. Finally, the dataset was organized into a model ready format, with a clear separation between input features and target variables. Overall, these preprocessing steps improved the quality of the data and ensured that the analysis and modeling results would be more accurate and reliable.

5. Exploratory Data Analysis (EDA)

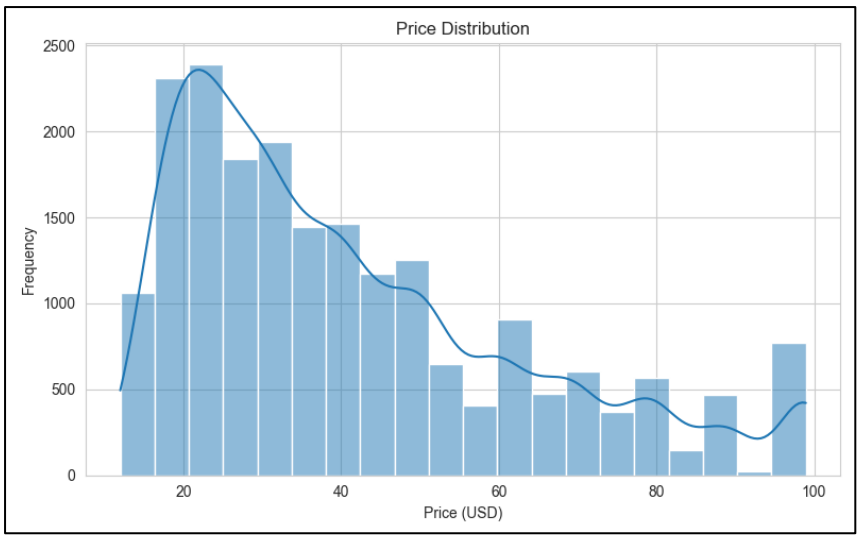
Exploratory Data Analysis (EDA) was conducted to understand Descriptive Statistics, patterns, relationships, and trends within the retail dataset. Various statistical summaries and visualizations were used to analyze key variables such as price, sales volume, promotions, seasonality, and product characteristics. The goal of this analysis is to generate meaningful insights that support business decision-making.

The descriptive statistical analysis provides a clear understanding of the key numerical variables in the dataset, including sales volume, price, and revenue estimates.

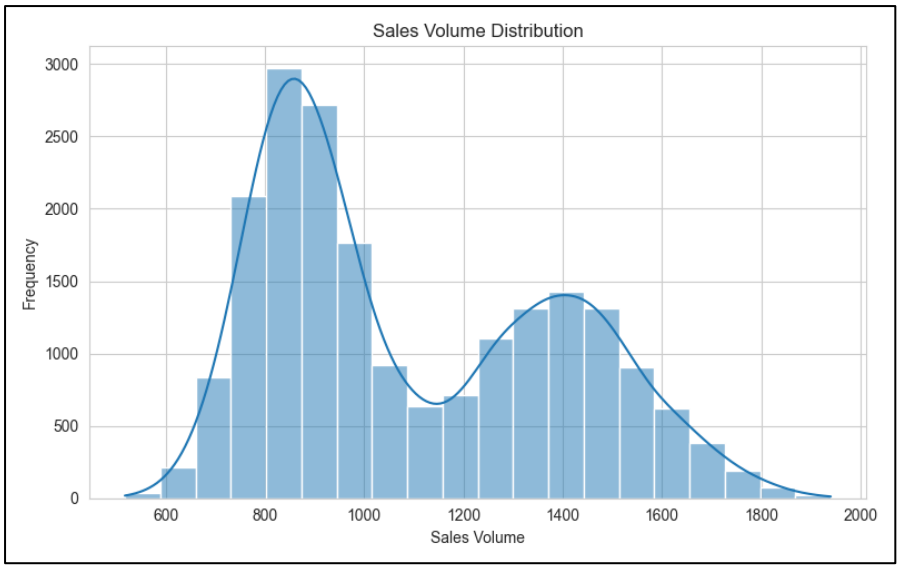
```
===== DESCRIPTIVE STATISTICS =====

          product_id  sales_volume         price  ...  revenue_estimate   name_length  description_length
count   20252.000000  20252.000000  20252.000000  ...      20252.000000  20252.000000        20252.000000
mean   208931.432303   1097.400454     41.545304  ...      43379.604548     35.460053          132.673711
std      8961.076507    298.234609     22.166818  ...      21972.583038      7.940394           51.307486
min    110075.000000    518.000000     12.000000  ...      11228.800000      9.000000           14.000000
25%    204442.750000    849.000000     23.950000  ...      26241.250000     30.000000           92.000000
50%    209505.500000    990.000000     35.950000  ...      38252.275000     35.000000          132.000000
75%    214568.250000   1364.250000     53.950000  ...      55709.150000     40.000000          168.000000
max    219631.000000   1940.000000     98.950000  ...     147039.700000     70.000000          313.000000

[8 rows x 8 columns]
```

The data reveals a diversified market strategy, where a high average sales demand (1,097 units) is successfully sustained across a broad pricing spectrum. However, the high revenue volatility (SD: >21,972) suggests that the business may be overly reliant on a few "superstar" products. With the Interquartile Range (23.95–53.95) proving that the mid-range segment is the most consistent driver of volume, there is a clear opportunity to optimize the product mix. Specifically, by narrowing the performance gap between the minimum (518) and maximum (1,940) sales values, the company can stabilize cash flow. Furthermore, the presence of premium products priced up to 98.95 indicates a healthy brand ceiling, suggesting that there is room to migrate mid-range customers toward higher-margin offerings through targeted marketing and enhanced product descriptions.

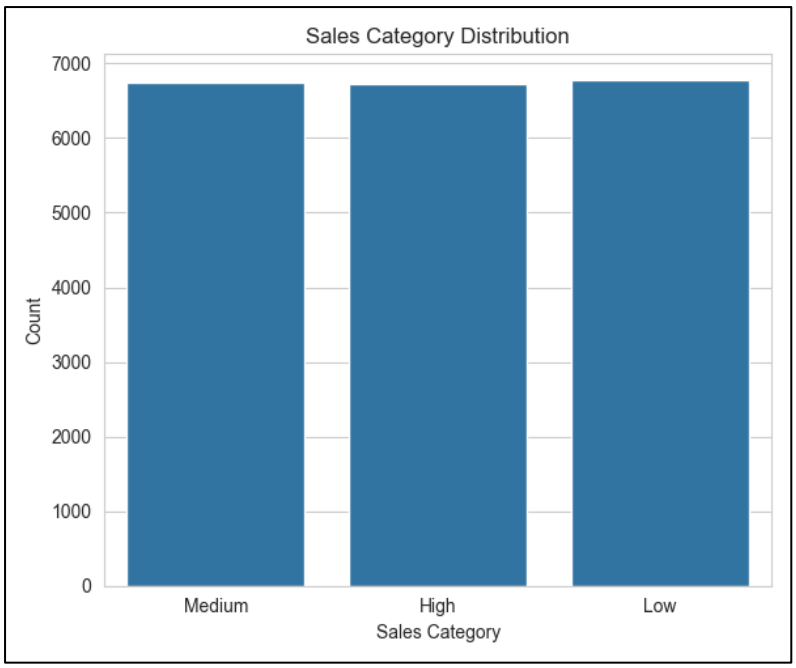
The distribution of product prices indicates that most items are concentrated within the lower to mid-range (approximately $15–$50), with only a small number of high-priced products. This right-skewed distribution suggests that the business primarily targets price-sensitive customers, focusing on affordability rather than premium pricing.



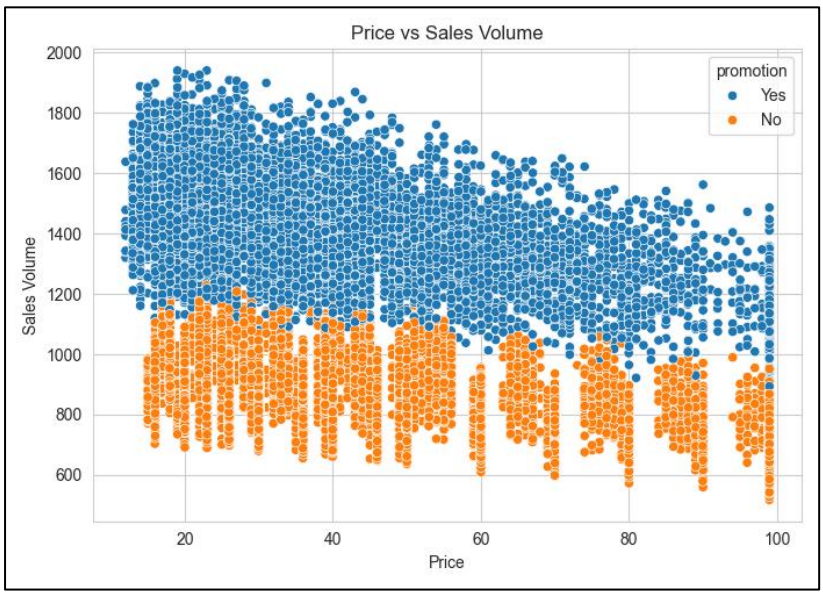
Similarly, the distribution of sales volume shows a bimodal pattern, with peaks around 800–1000 units and 1300–1500 units. This suggests the presence of two distinct product groups: moderately performing products and high-performing products. This distinction is useful for segmenting products based on demand levels.



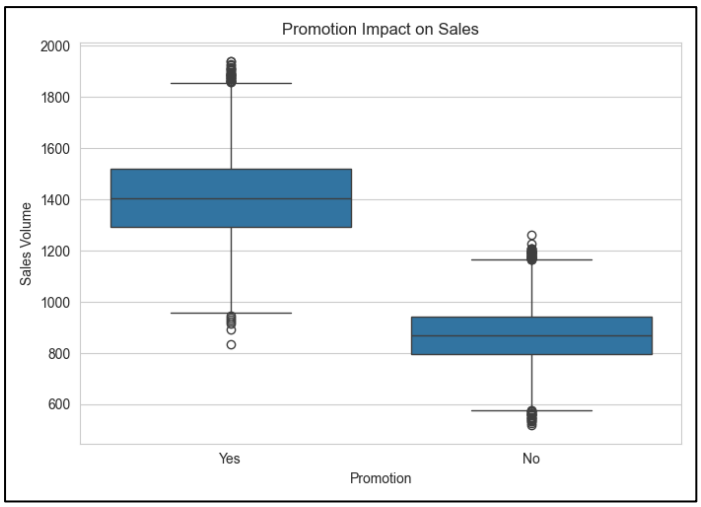
The analysis of sales categories reveals that products are almost evenly distributed among low, medium, and high sales groups. This balanced distribution indicates that the dataset is well-structured and suitable for further modeling and classification tasks.



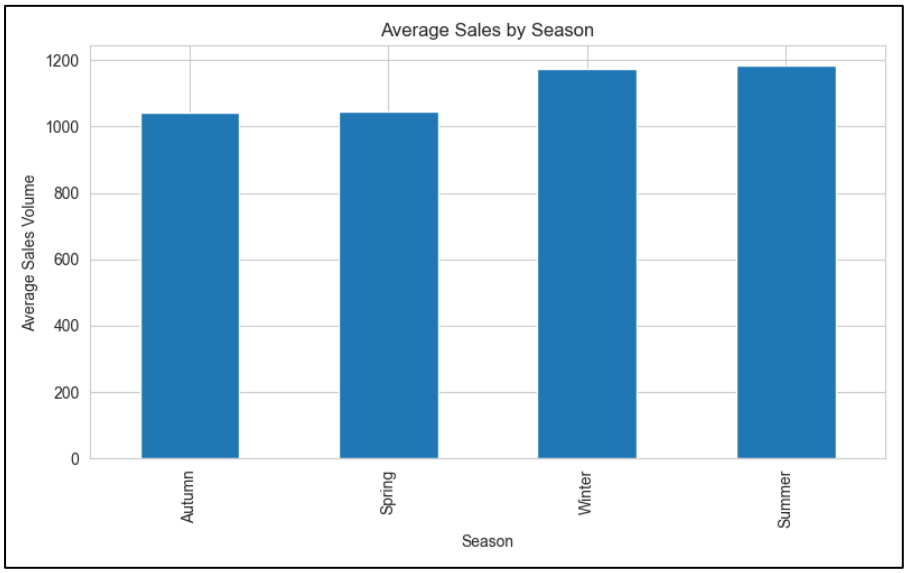
A key relationship observed in the analysis is between price and sales volume. The scatter plot shows a negative relationship, where higher-priced products tend to have lower sales volumes. This confirms that customer demand is highly price-sensitive. Additionally, products under promotion consistently achieve higher sales regardless of price level, highlighting the strong influence of promotional strategies.



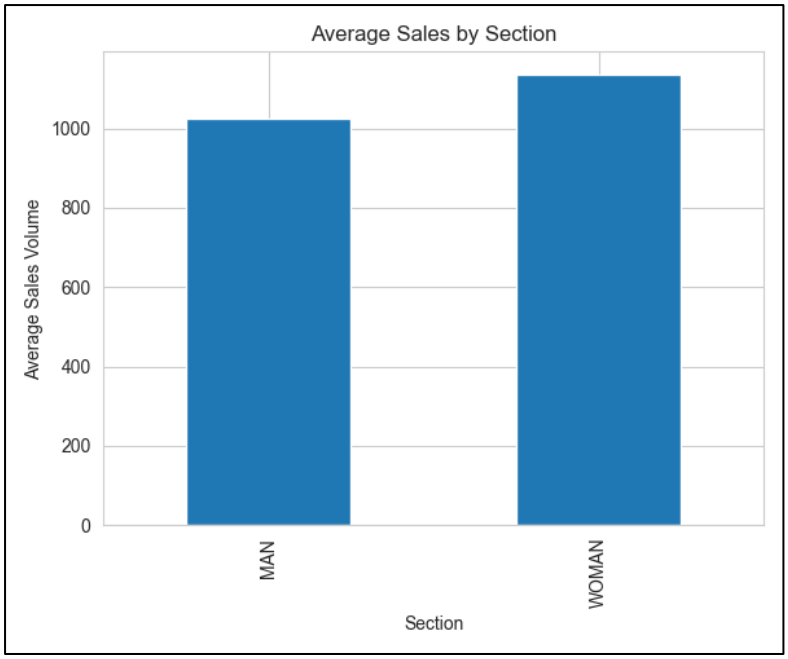
Further analysis of promotions using a boxplot shows that products with promotions have significantly higher median sales compared to those without promotions. This clearly demonstrates that promotional activities are a major driver of increased sales performance.



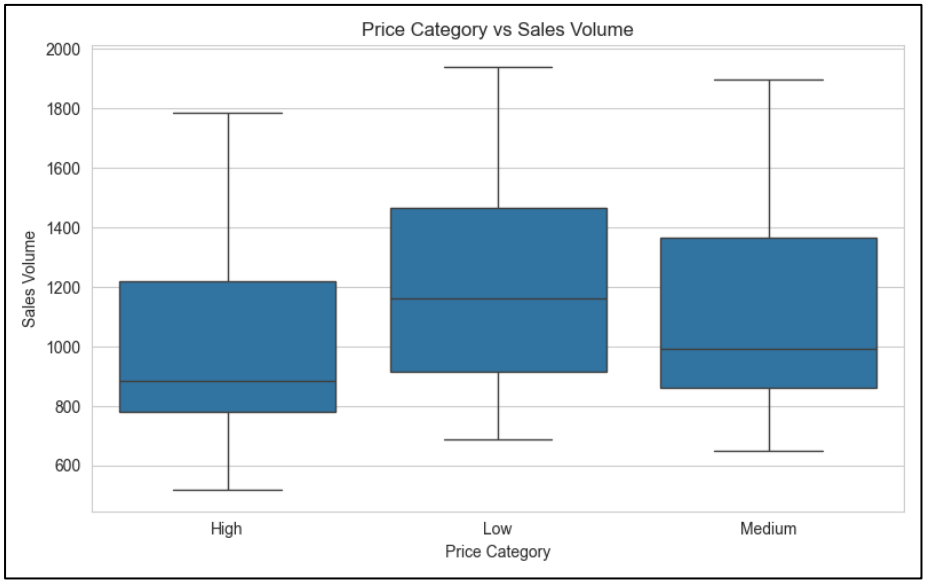
Seasonal analysis indicates that sales tend to be higher during Summer and Winter compared to Spring and Autumn. This suggests that consumer demand varies across seasons, likely influenced by weather conditions and seasonal trends in fashion retail.



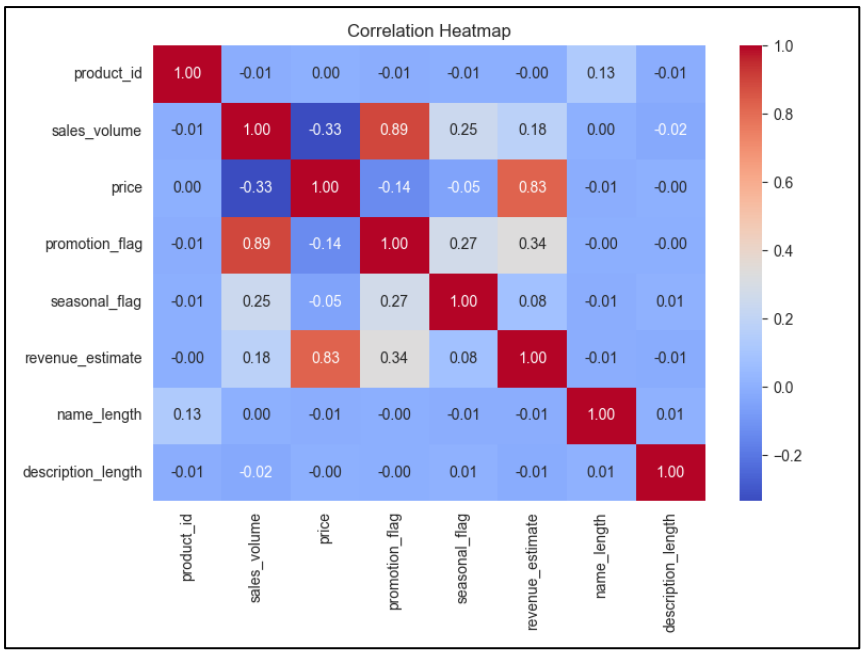
When analyzing product sections, it is observed that women's products generate slightly higher average sales compared to men's products. This implies stronger demand within the women's segment, making it a key area for business focus.



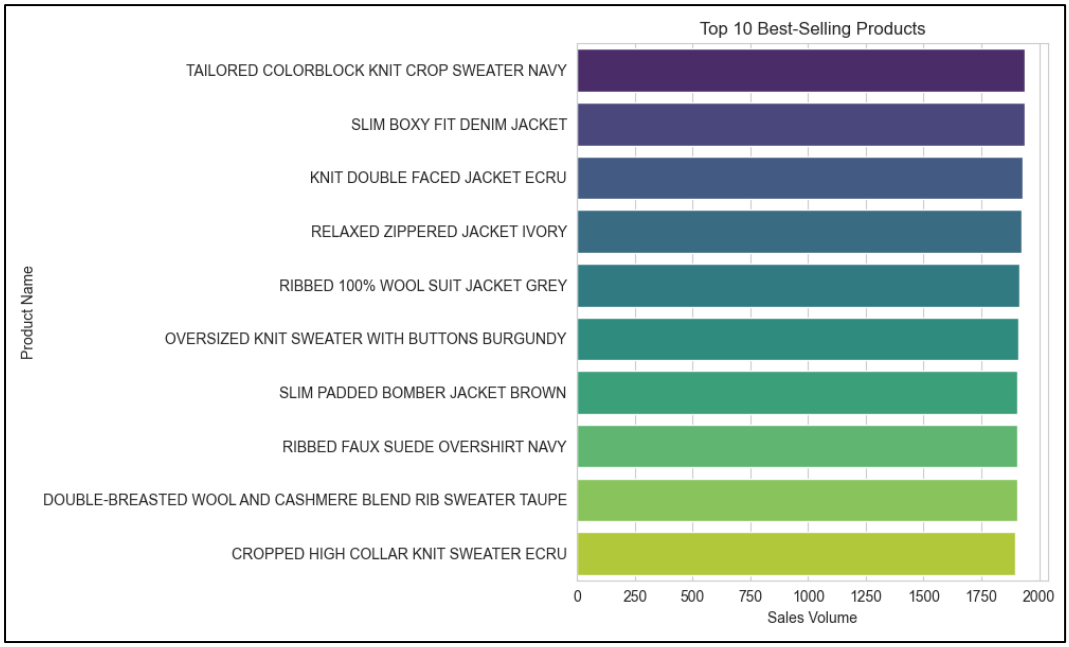
The relationship between price categories and sales volume further confirms that lower-priced products achieve higher sales compared to medium and high-priced items. This reinforces the conclusion that affordability plays a crucial role in driving demand.



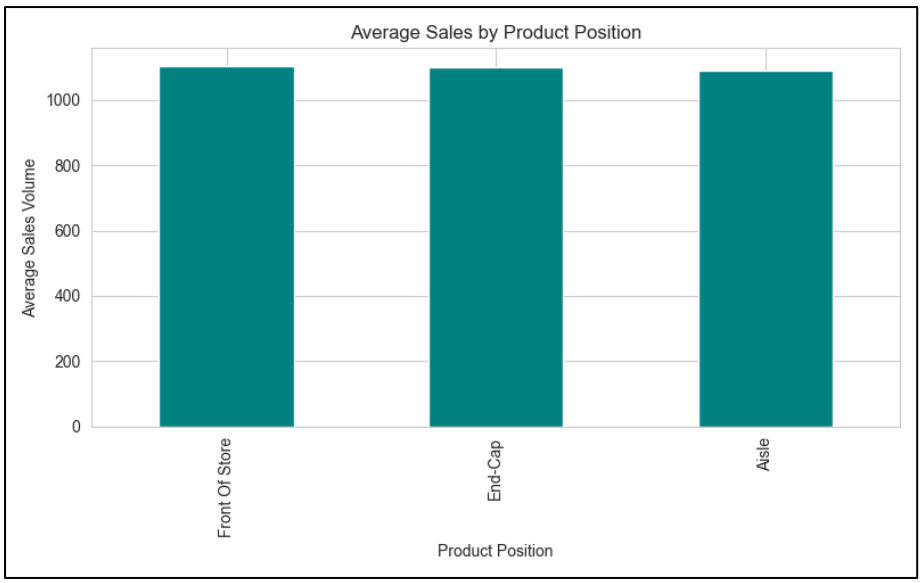
The correlation analysis provides additional insights into relationships between variables. A strong positive correlation is observed between promotion and sales volume, while price shows a moderate negative correlation with sales. Revenue, on the other hand, is strongly positively correlated with price. These relationships highlight the importance of balancing pricing and promotional strategies.



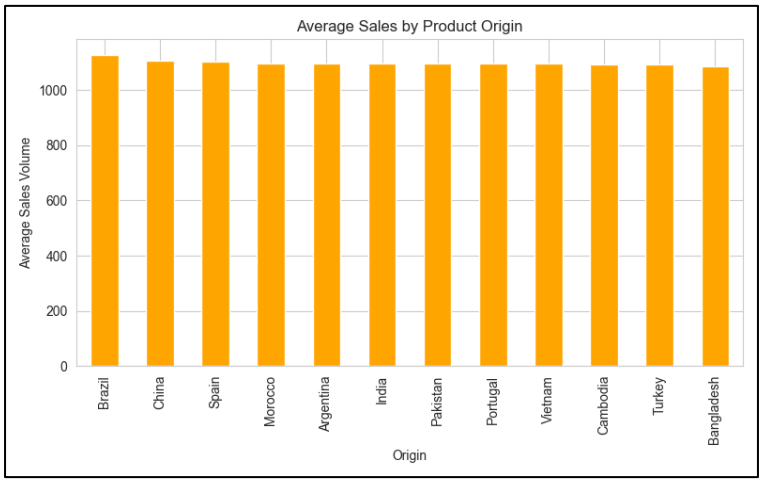
The analysis of top-selling products shows that a small number of products achieve very high sales volumes, close to 1900 units. These products likely benefit from optimal pricing, strong demand, and effective promotions. Identifying the characteristics of these products can help replicate their success across other items.



Further analysis of product positioning shows only minor differences in sales across store placements such as front of store, end cap, and aisle. This suggests that placement has a limited effect on sales compared to factors such as price and promotion.



Similarly, the analysis of product origin indicates that sales remain relatively consistent across different countries. This implies that product origin does not significantly influence customer purchasing decisions.



Overall, the EDA highlights that promotion, pricing, and seasonality are the most influential factors affecting sales performance. Promotions significantly boost sales, lower-priced products tend to perform better, and seasonal demand patterns exist. These findings provide valuable insights for optimizing pricing strategies, improving marketing campaigns, and managing inventory effectively.

6. Clustering

Clustering analysis was applied to organize similar products in the dataset using unsupervised learning, making it easier to group them based on factors like price, promotions, seasonality, and product attributes. This process helped uncover underlying patterns in demand and overall performance. In the next step, predictive modeling was used to estimate future sales through supervised learning, where the data was divided into training and testing sets and prepared using scaling and encoding techniques. Different models, such as Linear Regression and Random Forest, were trained and assessed using metrics like MAE, RMSE, and R^2, with the most accurate model selected based on the lowest error values. Overall, combining clustering with prediction offers both clear product grouping and reliable sales forecasts, helping businesses make smarter, data-driven decisions in retail.

6.1. Preprocessing

The clustering analysis grouped fashion retail products into distinct segments based on key features such as price, sales volume, promotions, seasonality, and product attributes like material and section. A key finding from this process is that effective data preprocessing played a critical role in producing meaningful and reliable clusters.

6.1.1. Numerical Feature Preprocessing

Handling missing values using the median helped reduce the impact of extreme values, while standardizing the data ensured that all features contributed equally to the clustering process. This prevented variables such as price from dominating the model and allowed more balanced grouping of products.

6.1.2. Categorical Feature Preprocessing

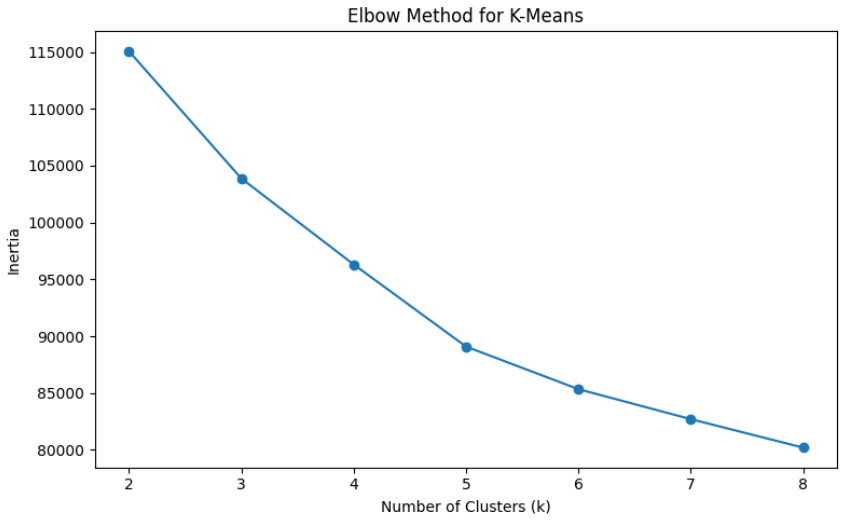
Replacing missing values with the most frequent category ensured data completeness, while one-hot encoding enabled non-numeric attributes to be included in the clustering. This allowed the model to capture both product characteristics and sales behavior, leading to more realistic segmentation.

As a result of these preprocessing steps, the clustering model clearly identified distinct product segments. For example, some clusters represented high-priced items with lower sales volumes, indicating premium or niche products. In contrast, other clusters included low-priced, high-demand items that are fast-moving in the market. Additionally, certain clusters showed a strong relationship between promotions and increased sales, highlighting the effectiveness of

discount strategies. Seasonal patterns were also observed, where specific products performed well only during certain periods.

6.2. Optimal Number of Clusters (Elbow Method)

The Elbow Method was used to determine the optimal number of clusters for the K-Means model by analyzing the relationship between the number of clusters (k) and inertia (within-cluster variance). From the plot, it can be observed that inertia decreases sharply as the number of clusters increases from 2 to around 4 or 5. After this point, the rate of decrease becomes more gradual, forming an "elbow" shape in the graph.



This indicates that adding more clusters beyond this point does not significantly improve the model performance. Based on the visual interpretation, the optimal number of clusters can be identified around k = 4, where a balance is achieved between minimizing inertia and avoiding unnecessary complexity.

6.3. Cluster Validation Using Silhouette Score

The selected model with $k = 4$ was evaluated using the Silhouette Score, which resulted in 0.1688. This indicates a moderate level of cluster separation, meaning the clusters are somewhat distinct but still have some overlap.

In general, a score closer to 1 shows well-separated clusters, while values near 0 suggest overlapping groups. In this case, the score reflects typical real-world retail data, where product features and demand patterns often share similarities across segments.

Despite this overlap, the clusters are still useful for identifying overall product patterns. Combined with the Elbow Method (which also suggested $k = 4$), this model provides a good

balance between simplicity and interpretability for business insights such as identifying high-performing or seasonal products.

6.4. Summarize

A cluster summary was generated to provide a clear, business-focused overview of the insights from each group.

```
===== CLUSTER SUMMARY =====
```

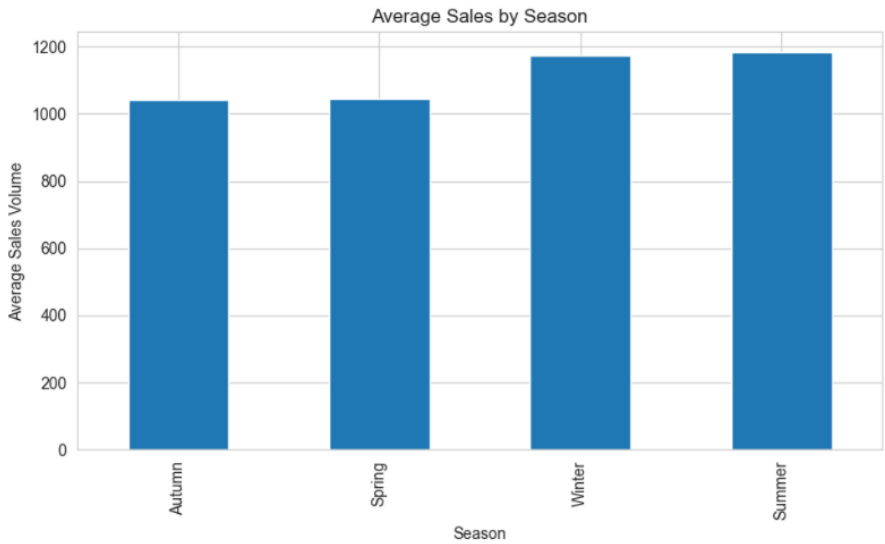
	cluster	avg_price	avg_sales_volume	avg_revenue	promotion_rate	seasonal_rate	product_count
0	0	44.23	871.58	37234.60	0.0	0.0	7272
1	1	37.58	1414.74	51618.73	1.0	1.0	5573
2	2	43.86	873.66	37060.76	0.0	1.0	4543
3	3	38.76	1408.19	52973.30	1.0	0.0	2864

The clustering analysis grouped products into four meaningful segments. Cluster 0, the largest with 7,272 items, consists of higher-priced, steady-selling products with no promotions or seasonal trends. Cluster 1 (5,573 products) captures lower-priced items with high sales, driven by both promotions and seasonality, while Cluster 2 (4,543 products) includes higher-priced, seasonal products that sell without promotions. Cluster 3, the smallest group with 2,864 products, features lower-priced, high-volume items reliant on promotions rather than seasonal demand. These clusters reveal clear patterns in pricing, sales, and demand drivers, providing actionable insights for marketing, inventory planning, and promotional strategies.

6.5. Visualizing

Certain graphs were plotted to visualize key patterns in the data, providing clearer insights into product performance, sales trends, and business opportunities.

This scatter plot visualizes product clusters by price and sales volume, making it easy to spot key patterns in the portfolio. It shows which clusters have high sales at low prices or low sales at high prices, helping identify growth opportunities or products that may need attention. The clustering highlights four distinct business patterns: high-priced, low-sales products (Cluster 0), low-priced, high-volume, promotion-driven items (Cluster 1), seasonal higher-priced products (Cluster 2), and moderate-priced, promotion-reliant items (Cluster 3). These insights provide actionable guidance for pricing, promotions, and inventory management, supporting more targeted business strategies.



This bar chart shows the average sales volume by season, making it easy to identify seasonal trends in the product portfolio. From the chart, we can see that sales are highest in Winter and Summer, while Autumn and Spring show slightly lower volumes. These insights

help businesses plan inventory, promotions, and marketing campaigns more effectively, ensuring that stock levels and sales efforts align with peak seasonal demand.

6.6. Business Insights from Clustering

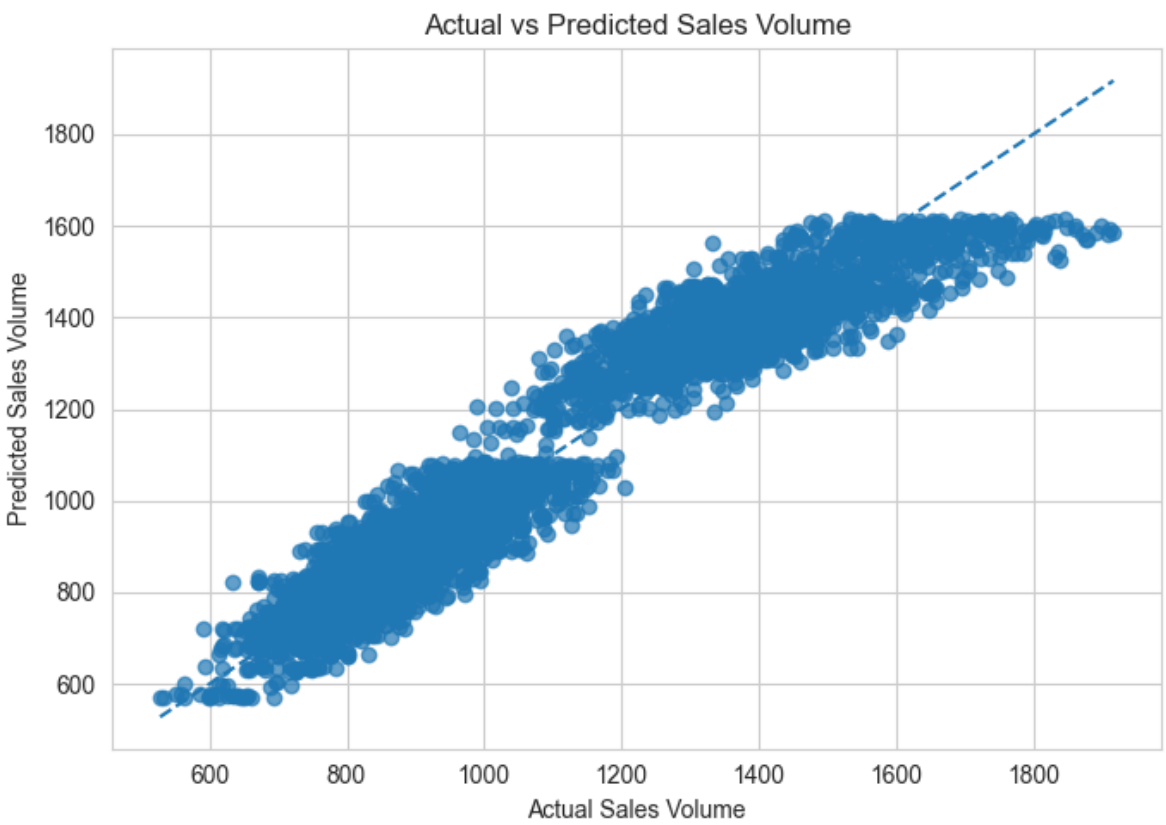
These clustering insights enable businesses to make data-driven decisions, enhancing efficiency and profitability. By understanding product segments, companies can optimize pricing, target promotions effectively, and manage inventory more strategically. High-demand products can be prioritized for restocking, slow-moving items can be promoted or reduced, and seasonal trends can guide timely stock planning. Overall, this approach shifts decision-making from intuition to analytics, driving increased sales, better resource allocation, and improved customer satisfaction.

7. Predictive Findings and Business Outcomes

To determine how different factors affect sales in fashion retail, we analysed predictive and future demand data to create useful predictions to guide business decisions. We focused on how pricing, promotions, and seasonality impact demand, and how these insights can help with planning and strategy.

7.1. Significant Findings

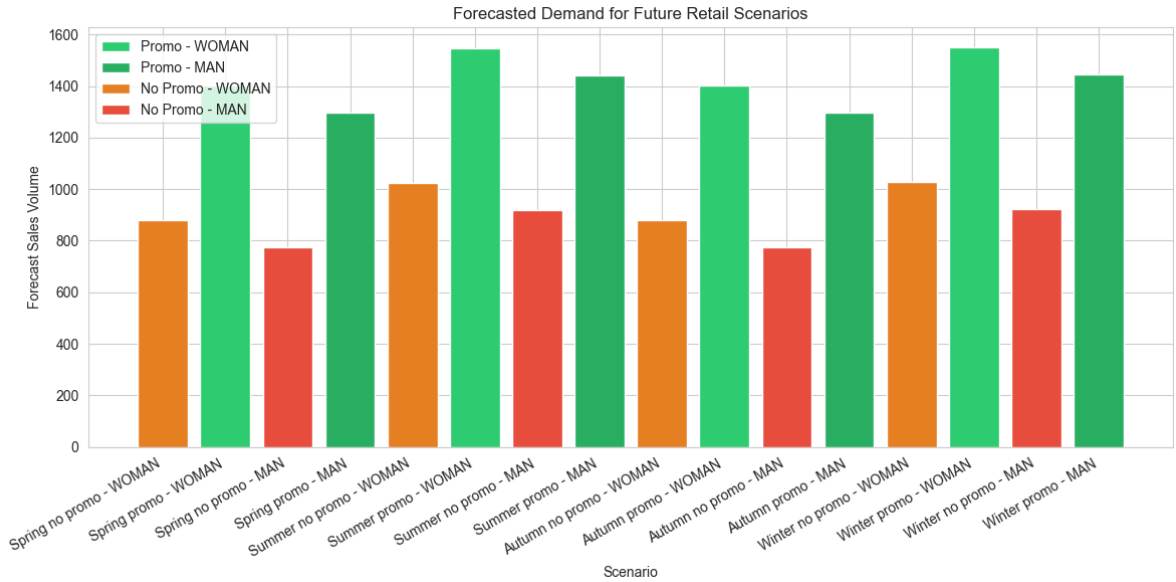
7.1.1. Model Performance and Reliability



The predictive model performs consistently, with the predicted sales volume closely matching actual results. The differences between predicted and actual values are small and show no major bias or instability. Most predictions fall within an acceptable range, supporting the model's reliability.

This behaviour is good enough for business use, particularly for forecasting and planning. While no model can predict demand perfectly, these outcomes provide a strong base for making informed decisions about future sales and operations.

7.1.2. Impact of Promotion



It is found that the promotions that have a big impact on sales. In many cases, demand increases significantly when promotions are used. Sales increase from less than 1,000 units to over 1,300 units with promotions.

These findings emphasize the importance of proper planning of promotions to stimulate short-term sales and make it remained competitive. Products with lower or mid-range prices sell more, while higher-priced items sell less unless they are promoted.

These results suggest that customers react strongly to price changes. However, pricing strategies alone do not always boost demand and work best when combined with promotions.

Pricing decisions should balance what customers want with the need to stay profitable.

7.1.3. Seasonal Influence

Seasonal trends affect demand, and their impact is stronger when combined with promotions. Forecasts show that demand is usually higher in Spring and Autumn.

Without promotions, the benefits of certain seasons are less noticeable. For example, demand is lower in Summer and Winter if there are no promotions. This means that seasonality alone does not drive demand, but it can influence other strategies to be more effective. Future demand forecasts help us understand how different conditions affect sales. When appropriate seasons and promotions are combined, demand can reach about 1400 units. When no promotion, especially in less favourable seasons, demand drops to around 900 units.

This large difference shows that promotions can greatly change expected sales. For planning, it is important to match marketing activities with seasonal trends to get the best results.

7.2. Business Outcomes and Recommendations

Taking into account of these findings, here are some practical takeaways: planned and targeted, especially during high-demand seasons like Spring and Autumn, to maximize impact.

Pricing strategies should remain competitive, balancing customer price sensitivity with the need for promotions to support higher-priced products, ensuring sufficient stock during busy periods, and avoiding excess inventory during low-demand periods.

The predictive model from this analysis can help make better, data-driven decisions in sales planning, marketing, and supply chain management.

7.3. Conclusion

Overall, the analysis shows that demand in fashion retail depends on several factors, with promotions and pricing as the main drivers. Seasonality also matters, but it works best when combined with promotions. Businesses can do better by using a smart approach to promotions and pricing, guided by data-driven insights. The predictive model gives a solid base for accurate forecasting and better alignment between business strategy and market trends.